

ACTIVITY 14 — LEARNER WORKSHEET**Descriptive Statistics and Implementation Planning**

TYPE	DURATION	DMAIC PHASE	MAPS TO
Elective Activity	45 min	CONTROL	A4, A5

Name	
Date	

What you are doing

Close the loop. Compute the measures of central tendency and dispersion that describe your data, then convert the selected countermeasures into a dated implementation plan with owners and barriers identified.

What you will produce: A descriptive statistics summary explaining the shape of the data and a dated implementation plan with owners and barriers.

Tools and techniques: Mean, median, range, standard deviation, implementation planning

The scenario

Contoso Service Desk handles employee IT requests. Employees complain that tickets take too long to be assigned, that they must keep chasing for status, and that agents apply different definitions of what counts as "resolved". Handoffs between the front line and the specialist queues are unclear, and although ticket data exists it is inconsistent. You are supporting the improvement team as a Yellow Belt — you do not lead the project.

Two weeks of data have been collected: 400 tickets, of which 96 contained at least one defect, with 120 defects recorded in total across 6 defect opportunities per ticket. Mean assignment time is 55 minutes (median 46) against a 30-minute improvement goal. The CSV files in this activity's data/ folder are that extract.

Files in this activity folder**data/ — the mock data you analyse:**

- assignment-times.csv — 400 rows (date, ticket_id, assignment_time_min)
- daily-summary.csv — 10 rows (date, tickets, mean_assignment_min, median_assignment_min, defects, missed_30min_goal)
- service-desk-tickets.csv — 400 rows (ticket_id, date, ticket_type, channel, assigned_queue, agent, assignment_time_min, met_30min_goal, defect_count, defect_categories, reopened)

templates/ — the worksheets you complete:

- descriptive-statistics.csv (statistic, formula_or_function, your_value)
- implementation-plan.csv (action, owner, start_date, due_date, barrier, mitigation)

Open these in Excel, Google Sheets or LibreOffice Calc. Work on a copy so the originals stay clean for revision.

data/assignment-times.csv is the raw column for the statistics.

Your steps

1. Compute the mean and median for your assignment-time data and compare them.
2. Compute the range and standard deviation to describe the spread.
3. Explain what the mean-versus-median difference reveals about outliers and skew.
4. Write the implementation plan: action, owner, date, barriers and mitigation.

Your working space

Complete every field. You may keep this worksheet and refer to it during the open-book assessment.

Step 1 — Compute the mean and median for your assignment-time data and compare them.

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Step 4 — Write the implementation plan: action, owner, date, barriers and mitigation.

Answers to the discussion prompts

1. Why do the mean and median differ in this data, and which would you report?

2. What does the standard deviation tell you that the mean cannot?

3. Which implementation action is most likely to slip, and what is the mitigation?

Reflect & discuss

Where does your workplace report an average that hides a skew?

What barrier most often stops improvements landing in your team?

Check your work

You can explain why the mean and median differ in your data, and every implementation action has an owner and a date.